Causation, Agency, and Existence: Structural Constraints and Interpretive Bridges

Sidong Liu, PhD
iBioStratix Ltd
sidongliu@hotmail.com

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Abstract

This essay examines the structural conditions under which agency, causation, and existence can be coherently discussed in the absence of foundational subsystem decompositions. Building on recent work in quantum information theory and algebraic approaches to quantum mechanics, Parts I–III develop a framework in which causal relations, agent boundaries, and existential claims emerge from stability properties of accessible observable algebras rather than from intrinsic substance or preferred factorizations.

Part I treats directional information asymmetry as an operational diagnostic that can support causal analysis when paired with explicit process, instrument, and intervention models, without requiring fundamental temporal ordering. Part II analyzes agency as boundary-stabilization—a non-scrambling subspace that propagates constraints without presupposing a metaphysically autonomous agent. Part III recasts existence in terms of relational form rather than intrinsic being, developing a notion of "emptiness" as absence of substance compatible with objectivity.

Part IV explores whether these structural features find formal parallels in Buddhist philosophical frameworks, particularly Yogācāra and Mādhyamaka traditions. The analysis emphasizes interpretive humility: parallels are offered as invitations to dialogue, not demonstrations of equivalence. The essay's contribution is methodological—clarifying structural constraints on emergence—rather than doctrinal.

Keywords: agency, emergence, structural realism, coarse-graining, accessible algebras, Buddhist philosophy, emptiness, comparative philosophy

Contents

1 Introduction **2**

2 Part I: Causation Without Foundations **3**
2.1 The Standard Picture and Its Assumptions 3
2.2 Directional Information Asymmetry 3
2.3 Thermodynamic Grounding 4
2.4 Counterfactuals Without Intrinsic Relata 4
2.5 Memory as Informational Constraint 4
2.6 Worked Example: Two-Qubit Directional Information 4
2.7 From Causation to Agency 5

3 Part II: Agency as Emergent Constraint Structure **6**
3.1 The Problem of Autonomous Agents 6
3.2 Agency as Boundary-Stabilization 6

3.3 Will as Constraint Propagation . . . 7
3.4 The Phenomenology-Structure Gap . . . 7
3.5 Degrees of Agency . . . 8
3.6 Relation to Free Will Debates . . . 8

4 Part III: Existence Without Substance 8
4.1 The Realism Problem . . . 8
4.2 Form Without Substance . . . 9
4.3 Emptiness as Technical Concept . . . 9
4.4 Examples of Relational Existence . . . 9
4.5 Structural Invariants and Objectivity . . . 10
4.6 Existence and Non-Existence . . . 10

5 Part IV: Interpretive Bridges to Buddhist Philosophy 11
5.1 Methodological Preface . . . 11
5.2 Yogācāra and Accessible Algebras . . . 11
5.3 Mādhyamaka and Emptiness . . . 12
5.4 Karma and Constraint Propagation . . . 13
5.5 Limits and Divergences . . . 14
5.5.1 Phenomenological vs. Structural Orientation . . . 14
5.5.2 Soteriological vs. Descriptive Goals . . . 14
5.5.3 Rebirth, Cosmology, and Ethics . . . 14
5.5.4 Ontological Commitments . . . 15
5.5.5 Methodological Incommensurability . . . 15
5.6 Interpretive Humility . . . 15

6 Conclusion: Structural Constraints and Interpretive Modesty 16
6.1 Methodological Contribution . . . 16
6.2 What This Essay Does Not Claim . . . 17
6.3 Open Questions . . . 17
6.4 Final Reflection . . . 17

1 Introduction

Contemporary philosophy of physics faces a structural puzzle. When quantum systems lack canonical subsystem decompositions, how should we understand causation, agency, and existence? If there is no preferred way to carve reality into parts, what becomes of the conceptual apparatus built on the assumption of well-defined relata?

This essay develops a framework in which these notions emerge from *stability properties of accessible algebras* rather than from intrinsic substance or preferred factorizations. The analysis proceeds in four parts.

Roadmap. Part I examines causation without temporal foundations, treating stable asymmetries in coarse-grained structure as operational diagnostics that can support causal analysis when paired with an intervention model. Part II analyzes agency as boundary-stabilization—the capacity of certain subsystems to maintain non-scrambling coherence while propagating constraints. Part III recasts existence in terms of relational patterns rather than intrinsic being, developing a technical sense of "emptiness" compatible with structural realism.

Part IV explores whether these structural features find formal parallels in Buddhist philosophical traditions, particularly Yogācāra and Mādhyamaka. The analysis emphasizes interpre-

tive humility: we identify structural similarities without claiming ontological identity, historical influence, or doctrinal convergence.

Methodological stance. This is a structural investigation, not a metaphysical proposal. We do not claim that consciousness creates reality, that Buddhist texts anticipated quantum mechanics, or that comparative philosophy resolves foundational problems. Rather, we clarify how certain structural constraints—particularly the absence of canonical factorization—reshape discussions of emergence, agency, and existence across different conceptual traditions.

The essay's contribution is methodological: it demonstrates how attention to algebraic structure can discipline interpretive claims and reveal unexpected points of contact between seemingly disparate frameworks.

Relation to prior work. This essay builds on the companion programme [1, 2]. Paper A establishes finite-dimensional channel separation under stated hypotheses and obtains metric inequivalence only conditionally on a reconstruction dictionary; Paper B treats stability criteria as heuristic diagnostics. The present essay explores possible philosophical consequences without upgrading those conditional inputs.

2 Part I: Causation Without Foundations

2.1 The Standard Picture and Its Assumptions

Causal relations are typically understood as relations between events ordered by time. Event A causes event B if: (i) A temporally precedes B, (ii) A and B are spatially connectible, and (iii) interventions on A counterfactually affect B [3].

This picture presupposes several structural features:

- A well-defined notion of temporal ordering
- Spatially localized events with clear boundaries
- Stable subsystem decompositions supporting counterfactual reasoning

In quantum contexts without canonical factorization, none of these features is guaranteed. Time may be emergent rather than fundamental [5]. Spatial locality depends on choice of coarse-graining [1]. Subsystem boundaries are algebra-dependent rather than intrinsic.

2.2 Directional Information Asymmetry

The former definition of causation by a bare cross-time quantum mutual information is withdrawn: standard quantum mechanics assigns no joint density operator to noncommuting events at two times without a process/instrument formalism. We retain a weaker, operational diagnostic.

Definition 2.1 (Instrument-relative directional information asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation, and specified instruments $\mathcal{I}_A, \mathcal{I}_B$ whose classical outcomes are $X_A^{t_i}, X_B^{t_i}$. The instruments and process define ordinary joint outcome distributions in separate forward and reverse experimental arrangements. Define*

$$\Delta^{\mathcal{I}}_{A\to B} := I\big(X_A^{t_1} : X_B^{t_2}\big) - I\big(X_B^{t_1} : X_A^{t_2}\big)\,, \tag{1}$$

*where both mutual informations are Shannon mutual informations of the declared classical outcome distributions. A positive value is a **directional informational asymmetry** relative to $(\mathcal{N}, \mathcal{I})$.*

Definition 2.1 does not define causation. It is instrument-, preparation-, and clock-order dependent and can arise from common causes or asymmetric noise. Quantum two-time objects such as pseudo-density matrices or conditional-state/process formalisms offer other explicit conventions [7, 8], but no convention-free expression $I(A_{\text{past}} : B_{\text{future}})$ is used here. A causal claim additionally requires an intervention comparison of the kind discussed in Section 2.4.

Protocol dependence and thermodynamic ordering. Different instruments, preparations, and coarse-grainings may produce distinct signs of $\Delta^{\mathcal{T}}_{A\to B}$. This is expected for an observational diagnostic and must not be reported as multiple fundamental causal orders.

Thermodynamic gradients may motivate which time ordering and instruments are physically salient [6], but entropy increase alone neither proves an intervention relation nor selects a unique coarse-graining.

2.3 Thermodynamic Grounding

The time labels used in Definition 2.1 may be aligned with a thermodynamic arrow in systems approaching equilibrium [6]. This supplies an ordering convention, not a derivation of causation from mutual information.

The thermodynamic arrow is coarse-graining dependent because macrostates are. Any causal conclusion must therefore state both the coarse-graining and the intervention model.

2.4 Counterfactuals Without Intrinsic Relata

Interventionist accounts of causation rely on counterfactual reasoning: A causes B if intervening on A would change B [4]. This appears to require well-defined intervention targets—subsystems A and B with stable identities.

Counterfactuals can be formulated relative to supplied subsystem algebras. An intervention on A is a declared quantum instrument/CPTP map that replaces the ordinary mechanism for A while holding the comparison protocol fixed. The causal quantity is then a difference between output statistics with and without that intervention, not the observational $\Delta^{\mathcal{T}}_{A\to B}$ alone.

This reformulation makes no reference to intrinsic subsystem boundaries. Intervention targets are defined by accessible algebras, which are themselves context-dependent.

2.5 Memory as Informational Constraint

Causal relations leave traces—memory records that constrain future accessible states. In quantum systems, memory can be understood as constraint propagation through entanglement structure [9].

A subsystem M acts as a memory of an instrument outcome $X_A^{t_1}$ if a later declared readout $X_M^{t_2}$ remains correlated with it despite environmental decoherence:

$$I(X_A^{t_1} : X_M^{t_2}) \gg I(X_A^{t_1} : X_E^{t_2}), \tag{2}$$

where all variables arise from the specified process and instruments.

Memory, on this account, is not storage of intrinsic properties but maintenance of relational structure across time—a pattern of correlations that persists under dynamical evolution.

2.6 Worked Example: Two-Qubit Directional Information

We compute Definition 2.1 in a valid two-qubit process. The example illustrates the diagnostic, not a discovery of causal direction from correlations.

Setup. At t_1, prepare a uniform classical bit in A and a fixed zero in B:

$$\rho_{AB}(t_1) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|10\rangle\langle 10|. \tag{3}$$

This matrix is positive and normalized, with spectrum $(1/2, 1/2, 0, 0)$. Use computational-basis instruments at both times.

Process. Let $\mathcal{N}_{t_1\to t_2}$ be the unitary CNOT channel with A as control and B as target. It maps $|x, 0\rangle$ to $|x, x\rangle$, so

$$\rho_{AB}(t_2) = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|, \tag{4}$$

again with spectrum $(1/2, 1/2, 0, 0)$. The relevant observable span

$$\mathrm{span}\{\mathbf{1}, \sigma_A^x, \sigma_A^z, \sigma_B^x, \sigma_B^z\}$$

is an operator subspace, not an algebra. If an algebra is required, one takes the unital $*$-algebra generated by these operators (here the full two-qubit matrix algebra).

Directional asymmetry. Writing the outcomes as bits, $X_A^{t_1}$ is uniform and $X_B^{t_2} = X_A^{t_1}$. Hence

$$I(X_A^{t_1} : X_B^{t_2}) = \log 2. \tag{5}$$

In the reverse arrangement, $X_B^{t_1} = 0$ is constant, so

$$I(X_B^{t_1} : X_A^{t_2}) = 0, \qquad \Delta_{A\to B}^{\mathcal{I}} = \log 2. \tag{6}$$

Interpretation. The sign is produced by the asymmetric preparation and the known controlled channel. It correctly detects directional information transmission in this protocol, but does not prove causation independently of the intervention/process specification.

Context-dependence. Changing the instruments, preparation, or channel can change or erase $\Delta_{A\to B}^{\mathcal{I}}$. The computed number is operationally real but protocol-relative.

2.7 From Causation to Agency

The transition from causation to agency requires an additional structural feature: *stable subsystem boundaries that support constraint propagation.*

An agent is not merely a locus of causal influence, but a subsystem capable of maintaining coherent constraint propagation despite environmental coupling. This suggests analyzing agency in terms of *non-scrambling subspaces*—degrees of freedom that resist rapid information delocalization [9].

Part II develops this connection, arguing that agency emerges from boundary-stabilization rather than metaphysical autonomy.

Summary. Directional information asymmetry is well defined only relative to a process and instruments. It may support a causal analysis when paired with interventions, but is not itself a definition of causation.

3 Part II: Agency as Emergent Constraint Structure

3.1 The Problem of Autonomous Agents

Traditional accounts treat agents as metaphysically autonomous entities—unified subjects possessing intrinsic intentionality and causal efficacy. This picture faces two challenges in quantum contexts without canonical factorization.

First, if subsystem boundaries are algebra-dependent, what distinguishes an "agent" from an arbitrary collection of degrees of freedom? Second, if quantum dynamics are unitary and deterministic at the global level, how can agents possess genuine causal autonomy?

We argue that both challenges dissolve once agency is recast as a structural feature—boundary-stabilization under constraint propagation—rather than a metaphysical primitive.

Physical grounding of algebra selection. Companion Paper B [2] supplies heuristic stability diagnostics relative to a candidate family, dynamics, fragmentation, and probe class. It does not derive a unique algebra from the Hamiltonian. We therefore take an agent candidate and its mereological boundary as model inputs below; their dynamical selection remains open.

3.2 Agency as Boundary-Stabilization

An *agent*, in the structural sense, is a subsystem whose boundaries remain stable under dynamical evolution and whose internal degrees of freedom exhibit coordinated constraint propagation.

Definition 3.1 (Agent-Like Subsystem)**.** *A subsystem $\mathcal{A}_{agent}$ exhibits **agent-like behavior** if:*

1. ***Boundary stability**: The subalgebra $\mathcal{A}_{agent}$ is approximately preserved under physically relevant dynamical maps:*

$$\sup_{0\le t\le T}\ \sup_{O\in \mathrm{Ball}(\mathcal{A}_{agent})} \|\mathcal{E}_t(O)-O\| \le \epsilon \tag{7}$$

   for timescales relevant to constraint propagation.

2. ***Post-transient scrambling control**: For a declared normalized probe class $V \in \mathfrak{V}$, observables in $\mathcal{A}_{agent}$ obey a rate or finite-window bound beyond the universal early-time regime, for example*

$$\sup_{O\in \mathrm{Ball}(\mathcal{A}_{agent}),\, V\in\mathfrak{V}} -\omega([O(t),V]^2) \le \delta \quad (t_0 \le t \le T), \tag{8}$$

   with $t_0 > 0$ specified by the model.

3. ***Constraint propagation**: Internal correlations support directed information flow toward system boundaries, enabling intervention on environmental degrees of freedom.*

Remark 3.2 (Operational Threshold for Meaningful Agency)**.** *Definition 3.1 is a structural modelling convention, not a derived biological threshold. Transient early-time commutator suppression does not establish sustained control.*

One may introduce a model-specific persistence ratio

$$R_{\mathrm{stab}} := \frac{\tau_{\mathrm{boundary}}}{\tau_{\mathrm{update}}}, \tag{9}$$

where τ_{boundary} is persistence of the effective control boundary and τ_{update} is its operational update time. No universal numerical cutoff for R_{stab} is known or claimed.

For biological systems these times are classical, coarse-grained control times, not quantum coherence times. Warm neural degrees of freedom decohere rapidly, and the present operator

language supplies no quantum-brain prediction [15]. Applying the structural analogy to organisms therefore requires an independently justified classical/open-system model.

This threshold distinguishes:

- ***Transient diagnostic satisfaction**: insufficient for operational agency*
- ***Sustained boundary stabilization**: a possible model input for persistent control*

Any empirical threshold is domain specific and must be calibrated rather than asserted from the formalism.

This definition makes no reference to consciousness, phenomenology, or intrinsic intentionality. Agency is characterized purely in terms of stability properties and information-theoretic structure.

3.3 Will as Constraint Propagation

What, then, becomes of "will" or "intention" in this framework?

We suggest that will can be understood as *constraint propagation structure*—patterns of internal correlation that bias future accessible states toward specific outcomes. An agent "wills" action A if its internal state ρ_{agent} is such that future measurements will register correlation with A with high probability.

Analogy: Thermostat control. Consider a thermostat coupled to a heating system. The thermostat's internal state (temperature reading) constrains future system behavior (heater activation), despite having no phenomenological experience. This is constraint propagation without metaphysical agency.

The analogy is limited: biological agents exhibit far richer constraint structure. But it clarifies the conceptual move—replacing metaphysical autonomy with structural analysis of how internal states bias future trajectories.

3.4 The Phenomenology-Structure Gap

An immediate objection: this account leaves no room for phenomenology—the felt quality of agency, the subjective sense of "I am acting."

We acknowledge this gap. The framework developed here is *structural*, concerned with information-theoretic organization. It does not address the *explanatory gap* between structure and experience [10, 11].

Speculative Connection: Neural Constraint Propagation

Caveat: The following connects structural features to neuroscience, but remains speculative. We note these as suggestive parallels, not established mechanisms.

Recent work in computational neuroscience suggests that neural "will" may correspond to hierarchical constraint propagation through cortical-basal ganglia loops [12]. Habitual behavior emerges when constraint patterns stabilize, while "voluntary" action involves flexible reconfiguration of these patterns [13].

If agency is boundary-stabilization, then the phenomenology of "willing" may correspond to proprioceptive monitoring of constraint reconfiguration—the felt sense of internal degrees of freedom reorganizing in preparation for action [14].

This remains highly speculative and does not bridge the explanatory gap. We raise it only to illustrate how structural analysis might interface with empirical research programs.

3.5 Degrees of Agency

The framework suggests that agency is not all-or-nothing, but admits degrees. Systems exhibit more or less agent-like behavior depending on:

- **Boundary stability duration**: How long does $\mathcal{A}_{\text{agent}}$ remain well-defined?
- **Scrambling timescale**: How quickly do internal correlations delocalize?
- **Constraint propagation fidelity**: How reliably do internal states bias future trajectories?

Simple thermostats exhibit minimal agency (short timescales, limited constraint structure). Biological organisms exhibit far richer agency (extended coherence, complex constraint networks). But both are comprehensible within the same structural framework.

3.6 Relation to Free Will Debates

Traditional free will debates ask: are agents causally autonomous, or are their actions determined by prior states and laws? This presupposes well-defined agent boundaries and unambiguous causal histories—precisely what the framework questions.

On the structural account, the relevant question is not "Are agents free?" but "Under what conditions do subsystems exhibit stable constraint-propagation structure?" This reformulation may dissolve certain traditional impasses while opening new empirical questions about stability conditions and scrambling timescales.

We do not claim to resolve free will debates—only to clarify how they depend on assumptions about subsystem structure that are themselves context-dependent.

Summary. Agency can be understood as boundary-stabilization supporting constraint propagation, rather than metaphysical autonomy. This account is eliminativist about intrinsic intentionality but realist about structural patterns of constraint. It leaves the phenomenology-structure gap unresolved but clarifies the information-theoretic conditions under which agent-like subsystems emerge.

4 Part III: Existence Without Substance

4.1 The Realism Problem

Parts I–II analyzed causation and agency as context-dependent but objective—dependent on coarse-graining structure but not on observer beliefs. This raises an existential question: if fundamental structures (subsystems, geometries, agents) are context-dependent, what ontological status do they possess?

Two extremes must be avoided:

1. **Naive realism**: Treating emergent structures as metaphysically fundamental, ignoring their context-dependence.
2. **Anti-realism**: Denying objective reality to context-dependent structures, collapsing into subjectivism.

We propose a middle path: *structural realism about relational patterns*. Existence is understood not as possession of intrinsic properties, but as participation in stable relational structures.

4.2 Form Without Substance

Consider the temperature of a gas. Temperature is *context-dependent*: it depends on which degrees of freedom are macroscopically accessible. Different coarse-grainings may yield different effective temperatures for the same microstate.

Yet temperature is not merely subjective. Given a coarse-graining, temperature is an objective, measurable quantity with predictive power. It is *relationally real*—real within a specified context, but lacking intrinsic existence independent of that context.

Definition 4.1 (Relational Existence)**.** *An entity X possesses **relational existence** relative to structure $\mathcal{S}$ if:*

1. *X is well-defined and stable within $\mathcal{S}$*
2. *X participates in objective relational patterns (correlations, symmetries, invariants)*
3. *X may be absent or differently constituted under alternative structures $\mathcal{S}'$*

This notion captures *form without substance*: patterns that are objectively real without possessing intrinsic being.

4.3 Emptiness as Technical Concept

We introduce *emptiness* as a technical term denoting absence of intrinsic existence compatible with relational reality.

Definition 4.2 (Emptiness (Technical Sense))**.** *An entity X is **empty** (in the technical sense) if:*

1. *X lacks intrinsic being independent of relational context*
2. *X exhibits stable patterns within specified contexts*
3. *The absence of intrinsic being does not entail non-existence or illusoriness*

This usage is stipulative and should not be confused with colloquial meanings ("containing nothing") or metaphysical nihilism ("nothing really exists"). Emptiness, in this technical sense, is compatible with robust realism about relational structures.

4.4 Examples of Relational Existence

Temperature. As discussed in §4.2, temperature is context-dependent but objective. Different coarse-grainings yield different effective temperatures, yet temperature remains a genuine physical quantity within each context.

Quasiparticles in condensed matter. Phonons, magnons, and other quasiparticles are collective excitations—emergent entities with no counterpart in the microscopic Hamiltonian. They are empty of intrinsic being (there are no "phonon particles" in fundamental theory) yet fully real within effective descriptions [16].

Operationally, quasiparticles are defined by stable relational patterns in measurable observables:

- **Spectral weight**: Well-defined peaks in momentum-resolved spectroscopy ($A(\mathbf{k}, \omega)$)
- **Dispersion relations**: Stable functional dependence $\omega(\mathbf{k})$ across parameter ranges
- **Finite lifetimes**: Decay rates $\Gamma(\mathbf{k})$ obeying systematic scaling laws

- **Scattering cross-sections**: Reproducible interaction amplitudes in transport experiments

These observables are context-dependent—they depend on temperature, pressure, doping, and measurement resolution—yet objectively real within specified experimental contexts. This exemplifies relational existence: patterns that are measurable, predictive, and stable, despite lacking intrinsic being independent of effective theory.

Subsystems in quantum mechanics. As established in prior work [1], subsystem decompositions are coarse-graining-dependent. A quantum state may admit infinitely many inequivalent factorizations, none privileged. Yet within any given factorization, subsystems exhibit objective entanglement structure and support meaningful predictions.

These examples illustrate a common pattern: entities that are *empty* (lacking intrinsic being) yet *existent* (participating in stable relational structures).

4.5 Structural Invariants and Objectivity

A potential objection: if everything is context-dependent, what grounds objectivity?

The answer lies in *structural invariants*—features preserved across context transformations. While subsystem decompositions are coarse-graining-dependent, certain global properties (total entropy, symmetry groups, topological invariants) remain well-defined independently of factorization.

Objectivity does not require context-independence. It requires only that relational patterns exhibit stability and predictive power within specified contexts, and that transformations between contexts preserve identifiable structural features.

This perspective aligns with *structural realism* in philosophy of science: what is objectively real is relational structure, not intrinsic substance [16, 17].

4.6 Existence and Non-Existence

The framework suggests a taxonomy of existential claims:

Type	**Characterization**	**Example**
Intrinsic existence	Context-independent being	Classical particles (if fundamental)
Relational existence	Context-dependent but objective	Temperature, subsystems
Conventional existence	Context-dependent and agent-relative	Money, legal rights
Non-existence	Absent from all relevant contexts	Phlogiston, luminiferous ether

Table 1: Taxonomy of existential claims. The framework developed here concerns relational existence—entities that are context-dependent but objective. These correspondences are analytical distinctions, not ontological commitments.

Emergent structures discussed in Parts I–II (causal relations, agent boundaries, geometric features) belong to the second category: relationally existent. They are empty of intrinsic being yet objectively real within specified coarse-graining contexts.

Summary. Existence can be understood in terms of relational patterns rather than intrinsic substance. "Emptiness," in the technical sense developed here, denotes absence of intrinsic being compatible with objectivity. This perspective avoids both naive realism and anti-realist eliminativism, offering a middle path grounded in structural invariance.

5 Part IV: Interpretive Bridges to Buddhist Philosophy

5.1 Methodological Preface

The structural features developed in Parts I–III—directional asymmetry as a causal diagnostic, agency as boundary-stabilization, existence as relational form—were derived independently of any particular metaphysical tradition. We now explore whether these features find formal parallels in Buddhist philosophical frameworks.

Several caveats are essential:

Methodological Caveats

1. **No historical causation**: We do not claim that Buddhist texts influenced quantum mechanics, or vice versa. Any parallels are structural convergences, not genealogical connections.
2. **No doctrinal advocacy**: Identifying formal similarities does not constitute endorsement of Buddhist metaphysics, soteriology, or religious practices.
3. **No cultural essentialism**: Buddhism comprises diverse traditions spanning two millennia. References here focus on specific textual traditions (primarily Yogācāra and Mādhyamaka), not "Buddhism" as a monolithic whole.
4. **No mystical reduction**: We reject interpretations conflating quantum mechanics with consciousness studies, New Age thought, or perennialist philosophy. Our analysis is structural and comparative, not mystical.
5. **Interpretive humility**: Parallels are offered as invitations to dialogue, not demonstrations of equivalence. The comparative exercise is exploratory, not conclusive.
6. **Naming is not corroboration**: The framework was deliberately named after ālaya-vijñāna. The parallel discussed below was therefore partly installed by the naming convention and cannot count as independent evidence for HAFF or for Yogācāra.

With these caveats in place, we proceed to examine possible structural correspondences.

5.2 Yogācāra and Accessible Algebras

Yogācāra ("Yoga practice") is a Mahāyāna Buddhist philosophical school emphasizing the role of consciousness (*vijñāna*) in constituting experienced reality. Central to Yogācāra is the concept of **ālaya-vijñāna** (Skt. *aalayavaija naaana*, "storehouse consciousness"), a foundational stratum of mind that stores karmic seeds (*bīja*) conditioning future experience [18, 19].

A structural parallel may be noted: ālaya-vijñāna functions as a holistic substrate from which individuated mental states emerge, analogous to how subsystem structures emerge from coarse-graining a global quantum state [21, 22].

Yogācāra Concept	HAFF Analog
Alaya-vijñāna (storehouse consciousness)	Global quantum state $\|\Psi_U\rangle$
Bīja (karmic seeds)	Eigenmodes of accessible algebras
Pravṛtti-vijñāna (active consciousness)	Effective subsystem ρ_{eff}
Āsraya-parāvṛtti (basis-transformation)	Coarse-graining map $\Phi_{\mathbf{c}}$

Table 2: Possible formal parallels between Yogācāra concepts and HAFF structures. **Disclaimer**: These correspondences are formal analogies highlighting structural similarities, not claims of ontological identity or historical influence. Yogācāra is a soteriological framework concerned with liberation from suffering; HAFF is a structural analysis of quantum emergence. The table illustrates conceptual resonances, not equivalences.

Critical Clarification: Against Panpsychism

The structural parallel between ālaya-vijñāna and the global quantum state $|\Psi_U\rangle$ does **not** imply:

- That quantum states possess consciousness or phenomenological properties
- That the universe is fundamentally mental or experiential (idealism)
- That matter is constituted by or reducible to mind (panpsychism)
- That information or quantum information is intrinsically conscious

The analogy is *purely structural*: both frameworks describe how individuated entities emerge from holistic substrates via coarse-graining or cognitive filtering. Yogācāra's substrate is explicitly mental (*vijñāna*, consciousness); HAFF's substrate is explicitly physical (quantum state in Hilbert space).
Any appearance of convergence concerns *formal pattern*—the structure of emergence—not ontological content. We emphatically reject interpretations that would construe this parallel as supporting quantum consciousness theories, New Age mysticism, or perennialist claims about universal mind.
The comparison is offered in the spirit of *structural analogy*, not metaphysical synthesis.

However, critical divergences must be noted:

- Yogācāra is primarily concerned with *mental* phenomena and the path to liberation, while HAFF analyzes *physical* structure without soteriological commitments.
- Ālaya-vijñāna is explicitly described as a form of consciousness, while $|\Psi_U\rangle$ is not imbued with phenomenological properties.
- The Yogācāra framework is embedded in Buddhist ethics and meditation practice, which have no counterpart in HAFF's purely structural analysis.

The parallel, if valid, concerns *formal structure*—both frameworks describe how individuated entities emerge from holistic substrates—not phenomenological or ontological content.

5.3 Mādhyamaka and Emptiness

The Mādhyamaka ("Middle Way") tradition, founded by Nāgārjuna (c. 2nd century CE), articulates **śūnyatā** (Skt. *shauunayataaa*, "emptiness") as the absence of **svabhāva** (Skt. *savabhaaava*, "intrinsic nature" or "own-being") [20].

This concept bears structural resemblance to the notion of "emptiness" developed in Part III (§4.3), where it denoted absence of intrinsic being without entailing non-existence. We emphasize that this parallel concerns the *structural form* of the claim—denial of substance while affirming functional reality—not historical influence or causal connection [23, 24].

Nāgārjuna's central argument proceeds via *prasaṅga* (reductio) reasoning: all phenomena are empty because they arise dependently (*pratītyasamutpāda*), and what arises dependently cannot possess intrinsic nature. This is formalized in the famous verse:

> *"Whatever arises dependently is said to be empty. That, being a dependent designation, is itself the middle way."* (MMK 24:18) [20]

A structural reading: entities lacking intrinsic being (empty) can nonetheless participate in stable relational networks (dependent arising). This maps onto the framework developed in §4.4: subsystems are empty (coarse-graining-dependent, lacking intrinsic factorization) yet existent (exhibiting objective entanglement structure).

Recursive emptiness and second-order structure. A subtle question arises: in Mādhyamaka, emptiness applies universally, including to emptiness itself—"emptiness is empty" (*śūnyatā-śūnyatā*). Does HAFF's relational existence admit similar recursive application?

The answer is affirmative in a formal sense. Selected algebras $\mathcal{A}_{\mathrm{sel},\mathbf{c}}$ are model data constrained by physical interaction structure, apparatus constraints, and resolution limitations. They are distinct from probe-effect, record, and fixed-point objects, and no context-free selection theorem is assumed.

Moreover, the coarse-graining maps $\Phi_{\mathbf{c}}$ that select accessible algebras are context-dependent: different experimental setups, measurement resolutions, or dynamical timescales induce different Φ structures. Thus, *the apparatus of emergence is itself emergent*—coarse-graining structure arises from prior coarse-graining choices in a potentially infinite regress.

This mirrors Mādhyamaka's insight that even the *tools of analysis* (concepts, language, logical operations) are empty—lacking intrinsic being while remaining functionally effective. In HAFF, even the "machinery" of accessible algebras is context-dependent, yet this does not undermine objectivity: structural invariants (entanglement entropy, symmetry groups) remain well-defined across contexts.

The parallel is formal: both frameworks acknowledge that *relational structure goes "all the way down,"* with no metaphysically foundational level immune to context-dependence. However, Mādhyamaka deploys this insight soteriologically (to undermine attachment to fixed views), while HAFF employs it descriptively (to clarify structural constraints on emergence).

Key divergence. Mādhyamaka śūnyatā is deployed soteriologically—to undermine attachment to fixed views and facilitate liberation. HAFF's "emptiness" is a technical descriptor of relational structure, with no soteriological function. The similarity is formal, not practical or existential.

5.4 Karma and Constraint Propagation

Buddhist karma doctrine holds that intentional actions leave traces (*saṁskāra*, "formations") that condition future experience. In Yogācāra, these traces are stored in ālaya-vijñāna as *vāsanā* ("karmic impressions") [22].

A structural analogy: karma may be understood as *constraint propagation through entanglement structure*—past actions (interventions on accessible algebras) leave informational imprints that bias future trajectories [25, 26].

This is consonant with the account of memory developed in §2.5: causal traces are not storage of intrinsic properties but maintenance of relational structure across time.

Critical limitation. Buddhist karma is inherently normative: actions are classified as wholesome (*kuśala*) or unwholesome (*akuśala*) based on ethical criteria and soteriological consequences. HAFF's constraint propagation is descriptive, lacking normative content. The formal parallel does not extend to ethical or soteriological dimensions.

5.5 Limits and Divergences

Having noted possible parallels, we now emphasize substantive divergences—areas where Buddhist frameworks and HAFF diverge structurally, methodologically, or conceptually. This section is weighted equally to §5.2–5.4 to prevent over-interpreting formal similarities.

5.5.1 Phenomenological vs. Structural Orientation

Buddhist philosophy is fundamentally concerned with first-person experience and liberation from suffering (*duḥkha*). Yogācāra and Mādhyamaka analyze consciousness, perception, and mental afflictions (*kleśa*) as prerequisites for soteriological transformation.

HAFF, by contrast, is a third-person structural framework with no phenomenological commitments. It analyzes information-theoretic organization without addressing subjective experience, qualia, or the explanatory gap between structure and consciousness.

Consequence: Any parallel between ālaya-vijñāna and $|\Psi_U\rangle$ cannot extend to phenomenological dimensions. HAFF does not explain consciousness, nor does it claim that quantum states possess mental properties.

5.5.2 Soteriological vs. Descriptive Goals

Buddhist frameworks are *pragmatic* in orientation: concepts are introduced to facilitate liberation, not to accurately describe metaphysical reality. As the Buddha reportedly stated, philosophical speculation is a "thicket of views" (*diṭṭhi-gahaṇa*) distracting from the path [27].

HAFF is *descriptive*: it aims to clarify structural constraints on emergence without prescribing practices or soteriological goals. There is no "path" in HAFF, no liberation to achieve, no suffering to overcome.

Consequence: Structural parallels do not imply that HAFF serves Buddhist soteriological purposes, nor that Buddhist practice requires acceptance of quantum mechanics.

5.5.3 Rebirth, Cosmology, and Ethics

Traditional Buddhist cosmology includes rebirth across multiple realms, karmic causation spanning lifetimes, and detailed ethical taxonomies. These elements are absent from—and irrelevant to—HAFF's structural analysis.

HAFF makes no claims about:

- Post-mortem continuity of consciousness
- Karmic retribution across lifetimes
- Ethical status of actions
- Cosmological realms or deities
- Meditation practices or contemplative attainments

Consequence: Identifying formal parallels does not validate Buddhist cosmology, rebirth doctrine, or ethical systems. The frameworks operate in disjoint conceptual spaces.

5.5.4 Ontological Commitments

While both frameworks reject intrinsic substance, they differ in ontological commitments:

- **Buddhist frameworks** (particularly Yogācāra) often privilege mind or consciousness as fundamental, with matter derivative.
- **HAFF** remains neutral on mind-matter relations, analyzing quantum structure without metaphysical commitments about consciousness.

Additionally, Mādhyamaka's "two truths" doctrine—distinguishing conventional (*saṁvṛti*) from ultimate (*paramārtha*) reality—has no clear analog in HAFF. HAFF distinguishes context-dependent from invariant structures, but this is epistemological (about what can be known) rather than ontological (about levels of reality).

5.5.5 Methodological Incommensurability

Buddhist philosophy employs contemplative introspection, textual hermeneutics, and dialectical reasoning as primary methods. HAFF employs mathematical formalism, operator algebras, and information theory.

These methodologies are not mutually translatable. One cannot *meditate* one's way to understanding quantum entanglement, nor *calculate* one's way to soteriological insight. The parallels identified are *structural*, not methodological.

Summary of divergences. The frameworks differ in:

1. Explanatory target (phenomenology vs. structure)
2. Pragmatic goal (liberation vs. description)
3. Scope (ethics/cosmology vs. physics)
4. Ontological commitments (consciousness-first vs. neutral)
5. Methodology (contemplative vs. mathematical)

These divergences are not defects but reflect different intellectual projects. Recognizing formal parallels does not collapse these distinctions.

5.6 Interpretive Humility

We conclude Part IV by reaffirming interpretive humility. The parallels identified—between accessible algebras and ālaya-vijñāna, between emptiness and śūnyatā, between constraint propagation and karma—are *suggestive but inconclusive*.

They suggest that:

1. Certain structural features (holistic substrates, relational existence, informational constraints) appear across traditions when thinkers grapple with similar conceptual problems.
2. Cross-cultural philosophical dialogue may benefit from precise structural comparison, avoiding both premature dismissal and uncritical conflation.
3. Formal parallels can motivate further investigation without requiring doctrinal convergence.

They do *not* suggest that:

1. Buddhist philosophy anticipated quantum mechanics or modern physics.

2. Quantum mechanics validates Buddhist metaphysics or soteriology.

3. Structural similarities entail ontological identity.

4. Comparative philosophy resolves foundational debates in either tradition.

The exercise is exploratory: we map conceptual terrain, noting points of contact and divergence, without claiming to adjudicate between frameworks. Our contribution is methodological—demonstrating how attention to algebraic structure can discipline comparative claims and prevent both over-interpretation and premature dismissal.

6 Conclusion: Structural Constraints and Interpretive Modesty

This essay has developed a structural framework for understanding causation, agency, and existence in quantum contexts without canonical subsystem decompositions. The analysis proceeded in four parts.

Part I: Causation. Directional information asymmetries can be defined relative to specified processes and instruments, without requiring fundamental temporal ordering or intrinsic relata. They may support causal analysis when paired with an explicit intervention model; thermodynamic gradients supply an ordering convention rather than a derivation of causation.

Part II: Agency. Agent-like behavior emerges from boundary-stabilization—subsystems maintaining non-scrambling coherence while propagating constraints. This account is eliminativist about intrinsic intentionality but realist about structural patterns. The phenomenology-structure gap remains unresolved.

Part III: Existence. Existential claims can be reformulated in terms of relational patterns rather than intrinsic substance. "Emptiness," in the technical sense developed here, denotes absence of intrinsic being compatible with objectivity. This perspective avoids both naive realism and anti-realist eliminativism.

Part IV: Interpretive Bridges. Formal parallels may exist between these structural features and concepts in Buddhist philosophy (particularly Yogācāra and Mādhyamaka traditions). However, substantive divergences in methodology, goals, and scope prevent conflation. Parallels are offered as invitations to dialogue, not demonstrations of equivalence.

6.1 Methodological Contribution

The essay's primary contribution is methodological: it demonstrates how structural analysis can discipline interpretive claims across multiple domains.

1. **In quantum foundations**: By clarifying how causation, agency, and existence depend on coarse-graining structure, the framework reveals which features are context-dependent and which admit invariant characterization.

2. **In philosophy of science**: By developing relational existence without metaphysical substance, the framework contributes to structural realist programs while avoiding reification of emergent entities.

3. **In comparative philosophy**: By identifying formal parallels while respecting substantive divergences, the framework models how cross-cultural comparison can proceed without cultural essentialism or premature synthesis.

6.2 What This Essay Does Not Claim

To prevent misreading, we reiterate what the essay does *not* claim:

- That consciousness creates reality or plays a fundamental physical role
- That Buddhist texts anticipated quantum mechanics or modern physics
- That quantum mechanics validates any particular metaphysical or religious tradition
- That structural parallels resolve foundational problems in physics or philosophy
- That comparative philosophy provides unique insights unavailable within traditions

The analysis is structural and comparative, not metaphysical or apologetic.

6.3 Open Questions

Several questions remain open:

1. **Phenomenology**: How, if at all, does structural organization relate to subjective experience? The framework developed here is silent on the explanatory gap.
2. **Normativity**: Can constraint propagation ground normative distinctions, or does ethics require additional conceptual resources beyond structural analysis?
3. **Comparative methodology**: What criteria should govern cross-cultural philosophical comparison? When do formal parallels indicate genuine convergence versus superficial similarity?
4. **Empirical implications**: Do different coarse-graining choices lead to observationally distinguishable predictions in realistic physical systems?
5. **Contemplative epistemology**: Can first-person contemplative methods contribute to structural understanding, or are mathematical and phenomenological investigations fundamentally disjoint?

These questions are not deficiencies but opportunities for future investigation. The framework provides conceptual scaffolding for pursuing them with greater precision.

6.4 Final Reflection

The absence of canonical subsystem decompositions in quantum mechanics is not merely a technical curiosity. It reshapes how we think about emergence, identity, and existence across multiple domains—from quantum gravity to philosophy of mind to cross-cultural hermeneutics.

By attending to structural constraints—particularly the dependence of effective descriptions on accessible algebras—we can navigate between naive realism and anti-realist eliminativism, between cultural essentialism and dismissive parochialism, between metaphysical dogmatism and interpretive nihilism.

The resulting picture is one of *structured pluralism*: multiple effective descriptions, none metaphysically privileged, yet constrained by objective relational patterns and transformation principles. Reality is not uniquely carved at the joints, but neither is it infinitely malleable. The joints themselves are context-dependent yet objective.

This perspective invites humility. We cannot claim unique access to fundamental structure, nor can we dismiss alternative frameworks as merely conventional. Instead, we map the space of

possibilities, identify structural invariants, and acknowledge the limits of any single descriptive framework.

In this spirit, the essay concludes not with answers but with refined questions—questions shaped by attention to algebraic structure, informed by cross-cultural comparison, and disciplined by interpretive modesty.

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